15 Open, High-Novelty & High-Impact Projects

1. Calibrated Early Risk Prediction with Uncertainty & Fairness Constraints

Problem Statement: Current disease risk models give only point estimates, are often poorly calibrated, and can be biased against demographic groups, making them unreliable for clinical decisions.

Existing System: Standard supervised models (Logistic Regression, XGBoost, simple Neural Nets) that output a single risk score without uncertainty or fairness guarantees.

Proposed System: A joint model that outputs calibrated risk probability + uncertainty interval + fairness audit flags, deployed as a FastAPI + Streamlit clinician decision-support tool.

2. Limited-Label Medical Image Classification with Active Learning + Edge-Ready Model

Problem Statement: Medical imaging AI needs large labeled datasets and heavy models, making it impractical for low-resource clinics and mobile devices.

Existing System: Fully supervised CNNs trained on large labeled datasets and deployed only on powerful servers.

Proposed System: Active-learning pipeline that selects the most informative samples + model compression (pruning/quantization) so the final model runs efficiently on edge devices with a simple web/mobile interface.

3. Fair Automated Short-Answer Scoring with Bias Auditing Dashboard

Problem Statement: Automated scoring systems can inherit and amplify demographic bias, and teachers have no easy way to monitor or correct this bias.

Existing System: Classical text classification models (TF-IDF + classifiers) that produce a score without fairness monitoring.

Proposed System: Fairness-aware scoring model + continuous bias-auditing dashboard that allows teachers to review, override, and log decisions for model improvement.

4. Spatio-Temporal Capacity Forecasting with Energy-Aware Recommendations

Problem Statement: Telecom operators forecast traffic but rarely optimize for energy consumption, leading to high operational costs and carbon footprint.

Existing System: Pure traffic forecasting models (ARIMA, LSTM, XGBoost) that ignore energy cost.

Proposed System: Spatio-temporal forecasting model that jointly optimizes load accuracy and energy, producing actionable recommendations (e.g., cell sleep modes) inside an operations dashboard.

5. Regulatory-Ready Credit Scoring with Monotonicity & Fairness Guards

Problem Statement: Credit scoring models must be accurate, interpretable, monotonic, and fair to meet regulatory requirements, but most ML models violate these constraints.

Existing System: Black-box or unconstrained models that achieve high AUC but fail monotonicity and fairness checks.

Proposed System: Constrained model (monotonic + fairness regularized) that produces score + reason codes + continuous fairness monitoring, packaged as an underwriter decision tool.

6. Cost-Sensitive Fraud Detection with Operator Feedback Loop

Problem Statement: Fraud detection systems optimize accuracy or F1, but real cost of false positives vs false negatives is ignored, and operator feedback is rarely used to improve the model.

Existing System: Imbalanced classification models (XGBoost, Isolation Forest) evaluated only on PR-AUC or F1.

Proposed System: Cost-sensitive model with explicit FP/FN cost matrix + live operator console that allows labeling and feeds back into continuous model improvement.

7. Continual Learning under Concept Drift with Production Monitoring

Problem Statement: Production models degrade silently when data distribution drifts, and there is no systematic way to detect drift and adapt while preserving past knowledge.

Existing System: Static models retrained periodically from scratch or simple online learners without monitoring.

Proposed System: Continual learning system with drift detection, controlled adaptation, forgetting metrics, and a live monitoring dashboard that visualizes performance over time.

8. Uncertainty-Aware Models with Safety Gate for High-Stakes Decisions

Problem Statement: High-stakes systems (medical, financial, legal) make predictions even when the model is uncertain, leading to dangerous errors.

Existing System: Standard models that always output a prediction without a reliable confidence mechanism or reject option.

Proposed System: Calibrated uncertainty model with an explicit safety gate (reject option) that escalates low-confidence cases to a human review queue via a decision-support interface.

9. Explainable Graph ML for Relationship Risk + Investigation UI

Problem Statement: Risky relationships (fraud rings, supply-chain risk, patient similarity) are hard to detect and even harder to explain to investigators.

Existing System: Tabular models or basic graph algorithms that give a risk score without clear explanations.

Proposed System: Graph-based risk model with edge/feature attributions + interactive investigation console that visualizes the graph and explanations.

10. Physics-Informed Neural Networks for Real-Time Wearable Physiological Forecasting

Problem Statement: Pure data-driven models for wearable signals ignore known physiological constraints and are rarely optimized for edge deployment.

Existing System: Standard LSTM/CNN time-series models trained only on data.

Proposed System: Physics-informed neural network that respects physiological laws, provides uncertainty estimates, and runs efficiently on edge devices with a real-time dashboard.

11. Multi-View Contrastive Learning for Robust Sensor Fault Detection under Distribution Shift

Problem Statement: Sensor fault detectors trained on one operating condition fail when the environment or machine behavior changes.

Existing System: Supervised or simple unsupervised anomaly detectors that assume stationary data.

Proposed System: Multi-view contrastive learning model that learns robust representations and includes live shift detection + operator alerting dashboard.

12. Causal Feature Attribution for High-Stakes Tabular Decisions with Actionable Recourse

Problem Statement: Current explainability tools (SHAP, LIME) show correlations, not causal effects, and do not tell users what they can realistically change.

Existing System: Correlational feature attribution methods without recourse generation.

Proposed System: Causal attribution model that identifies true drivers and generates realistic, actionable recourse suggestions inside a decision-support interface.

13. Hierarchical Time-Series Forecasting with Cross-Series Reconciliation and Uncertainty

Problem Statement: Forecasts at different levels (SKU, category, total) are inconsistent, and most systems provide only point forecasts without reliable uncertainty.

Existing System: Independent forecasting models at each level without reconciliation or proper probabilistic outputs.

Proposed System: Hierarchical forecasting system with cross-series reconciliation, calibrated probabilistic forecasts, and a planning dashboard for inventory/operations decisions.

14. Graph-Based Early Warning System for Cascading Failures in Critical Infrastructure

Problem Statement: Failures in power grids, telecom networks, or transportation systems can cascade, but current systems detect local failures too late and without clear explanations.

Existing System: Local anomaly detectors or simple threshold-based alarms.

Proposed System: Graph-based early-warning model that predicts cascade risk with lead time, provides explanations, and visualizes risk on an interactive network dashboard.

15. Fairness-Aware Continual Learning for Evolving Demographic Distributions

Problem Statement: Models that are fair at deployment time become unfair as the demographic distribution of the population drifts over months or years.

Existing System: Static fairness-aware models or continual learning models that ignore fairness over time.

Proposed System: Continual learning framework that jointly maintains accuracy and fairness under demographic drift, with a long-term fairness monitoring dashboard.